DIGITAL ASSIGNMENT 3

Final Analysis, Interactive Dashboard and Project Conclusion

Customer Segmentation and Churn Prediction | Online Retail II

Programming for Data Science (BCSE207L)

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01 STARTING POINT

DA2 ended with two open questions

1

Which model is actually best?

The two leading churn models were separated by 0.0016 of ROC-AUC on a single train-test split. One McNemar test on one split is thin evidence for a decision.

2

What does a business do with it?

Churn risk, high-value status and segment were three separate outputs. None of them tells anyone what to do about a particular customer.

CARRIED FORWARD FROM DA2

779,425

valid transactions

5,281

customer records

24

models trained

0.80

ROC-AUC ceiling

02 SCOPE

Four deliverables in this phase

1 Statistical validation

Four independent procedures applied to the same five models, instead of one test on one split.

2 Consolidated scoring

All 5,281 customers scored into one record: risk band, value tier, segment and a recommended action.

3 Two dashboards

A self-contained Plotly HTML file and a reactive R Shiny application, reading the same exported data.

4 Interpretation and close

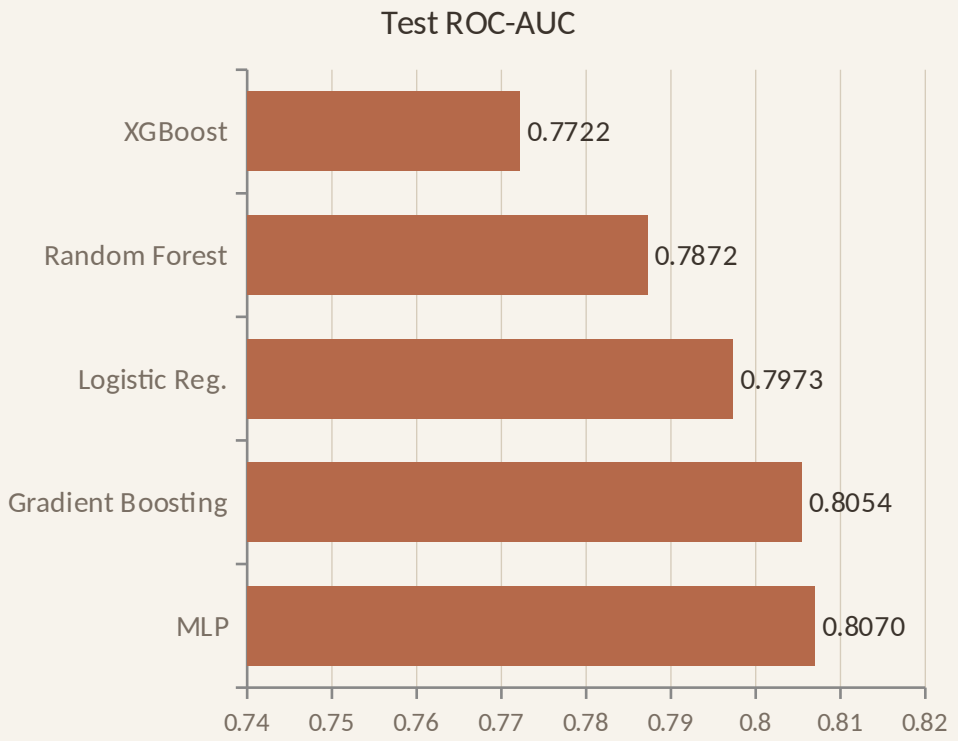
What the results mean for a business, where they stop being trustworthy, and what comes next.

Five models, one identical test bed

Same twelve features, same stratified 75 / 25 split of 3,960 training and 1,321 test customers, and the same five cross-validation folds for every model. Identical folds are what make the paired tests valid.

Model	Test ROC-AUC	Mean CV	Recall	Accuracy
Neural Network (MLP)	**0.8070**	**0.7964**	0.8021	0.7441
Gradient Boosting	0.8054	0.7960	**0.8088**	0.7282
Logistic Regression	0.7973	0.7879	0.7821	0.7366
Random Forest	0.7872	0.7823	0.7901	0.7222
XGBoost	0.7722	0.7606	0.7741	0.7055

The ordering reproduces DA2 exactly. But the gap between the top two is 0.0016 - smaller than the spread between folds.



04 METHOD

One test cannot settle a 0.0016 gap

1

Confidence intervals

95% t-intervals on the mean fold score. Describes the range the true performance likely sits in.

2

Paired t-test and Wilcoxon

Every model on identical folds, so differences are paired and testable directly. All ten pairs.

3

Pairwise McNemar

Looks only at the customers where two models disagree, and asks whether the disagreement is lopsided.

4

Bootstrap intervals

1,000 resamples of the test set with replacement. Assumes no distribution at all.

Four procedures that answer different questions. Where they agree, the conclusion is safe. Where they disagree, the disagreement is itself the finding.

How wide is the uncertainty around each score?

95% CONFIDENCE INTERVAL ON MEAN FOLD SCORE

Model	Lower	Upper
Neural Network	0.7856	0.8071
Gradient Boosting	0.7847	0.8074
Logistic Regression	0.7752	0.8006
Random Forest	0.7704	0.7942
XGBoost	**0.7481**	**0.7731**

The top two intervals overlap almost completely, so the two cannot be separated. The XGBoost interval overlaps with no other model, which makes it the clear weakest of the five.

BOOTSTRAP, 1,000 RESAMPLES OF THE TEST SET

0.045
typical interval width

0.0016
gap between top two

- Every bootstrap interval is roughly 28 times wider than the gap it would have to resolve
- Read alone, the bootstrap says the single-split ranking is not a firm ordering at the top
- This is the honest reading of a leaderboard built on one split

06 SIGNIFICANCE

Two valid tests, two opposite verdicts

PAIRED t-TEST ACROSS FOLDS

8 of 10

pairs significant at the 5% level

Neural Network vs Gradient Boosting
mean difference 0.0003 p = 0.8190 not significant

Logistic Regression vs Random Forest
p = 0.0852 not significant

Wilcoxon cannot go below p = 0.0625 with five folds, so it supports rather than decides.

McNEMAR ON THE TEST SET

p = 0.0402

Neural Network vs Gradient Boosting - significant

58 customers MLP gets right that GB misses

37 customers GB gets right that MLP misses

Only the off-diagonal counts matter: the 1,226 customers both models agree on carry no information about which is better.

07 RECONCILIATION

Why both tests are right

	Paired t-test	McNemar
What it compares	Average ranking quality (ROC-AUC)	Hard 0/1 predictions at a fixed threshold
Measured across	Five different partitions of the data	One fixed set of 1,321 customers
Question it asks	Would it rank better on a fresh sample?	Do they misclassify the same people?
Verdict here	Equivalent, p = 0.8190	Different, p = 0.0402

The Neural Network makes measurably different errors on this test set, but across resampled folds its ranking ability is no better. The overlapping bootstrap intervals support both readings at once.

SO WHICH ONE SHIPPED, AND WHY

Highest recall

0.8088 of five models - a missed churner is the expensive error

Already tuned

Grid search and threshold 0.453 lifted recall to 0.873 in DA2

Cheaper to retrain

Faster and more stable, with fewer settings that affect results

Three model outputs become one actionable record

Every one of the 5,281 customers scored with the tuned Gradient Boosting model at the optimised threshold of 0.453, then given three derived fields.

Risk band — Low below 0.3 | Medium 0.3 to 0.6 | High above 0.6

Value tier — Quartiles of lifetime spend: Bronze, Silver, Gold, Platinum

Revenue at risk — Lifetime spend multiplied by churn probability

Revenue at risk is the field that makes the view useful: it reorders the list so a GBP 50,000 customer at 60% outranks a GBP 200 customer at 90%.

TOTAL EXPOSURE ACROSS ALL CUSTOMERS

GBP 3.67M

revenue at risk

GBP 13.93M

total lifetime value

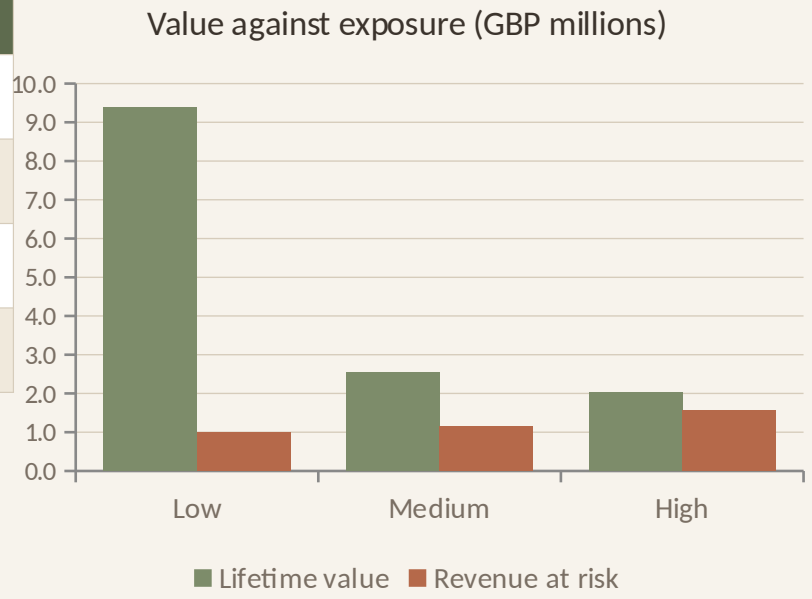
26%

of customer value exposed

Largest single account: customer 12346, GBP 77,556 of spend at probability 0.902, placing GBP 69,979 at risk.

The money and the risk sit in different places

Risk band	Customers	Avg. probability	Lifetime value	Revenue at risk
Low (below 0.3)	989	0.154	9,371,701	976,223
Medium (0.3 to 0.6)	1,677	0.465	2,539,303	1,135,783
High (above 0.6)	2,615	0.787	2,019,552	**1,556,128**
Total	**5,281**	**0.566**	**13,930,556**	**3,668,134**



The central insight of the project

Most of the money - GBP 9.37M - sits with the 989 low-risk customers, who are going nowhere. The high-risk band holds far less value at GBP 2.02M, but because those customers are likely to leave, their weighted exposure of GBP 1.56M is the largest of the three. Ranking by probability alone sends the budget to the wrong people.

Risk and value together decide the action

	Gold or Platinum	Bronze or Silver
High risk	**Urgent retention call**	**Automated win-back offer**
Medium risk	Proactive engagement	Monitor
Low risk	Loyalty / upsell	Maintain

The logic is deliberately simple so it can be explained to a non-technical reader and changed without retraining anything.

WHERE THE EXPOSURE LANDS

569

urgent retention calls

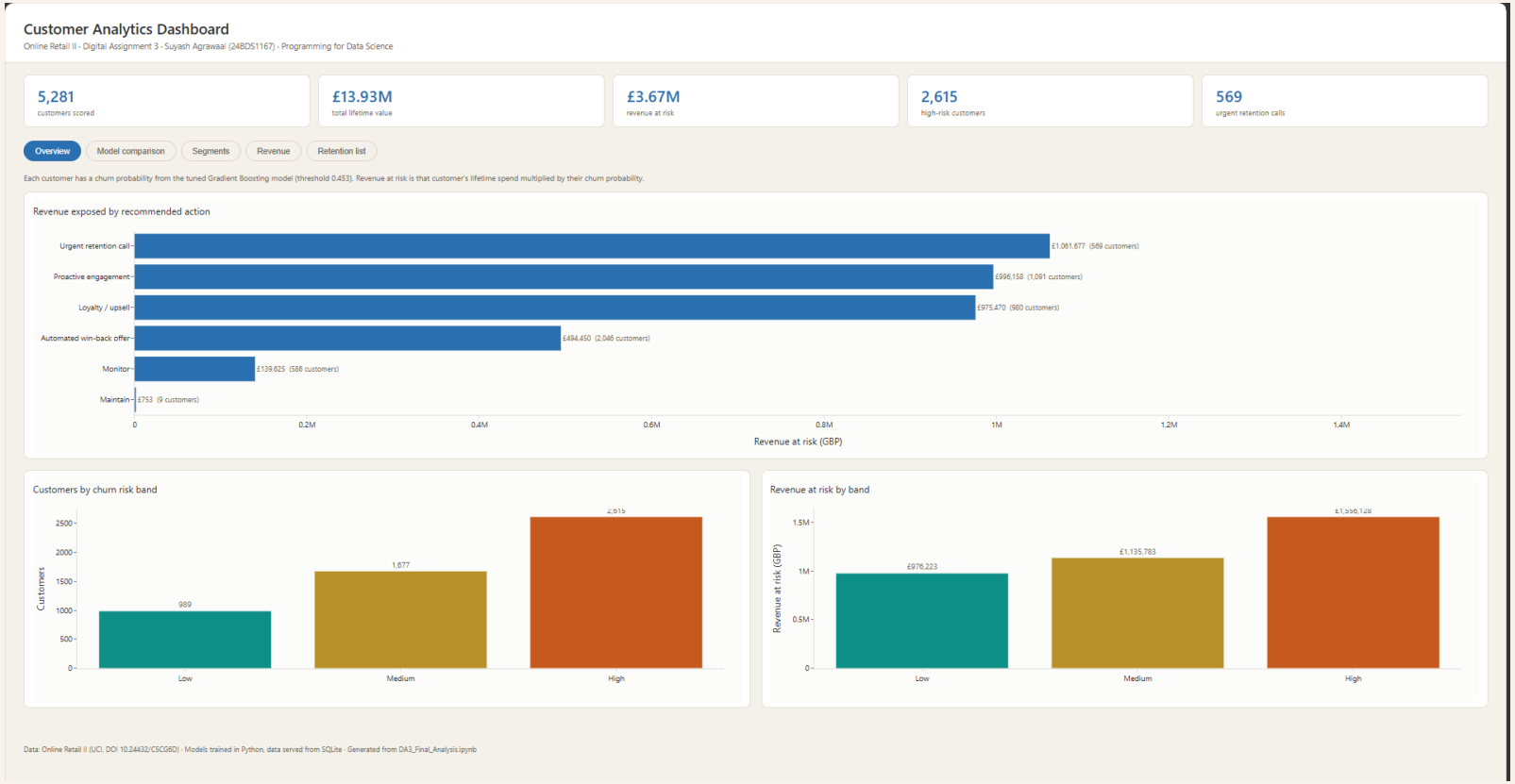
carrying GBP 1,061,677 of revenue at risk - almost a third of the total, which justifies individual human contact.

2,046

automated win-back offers

highest average risk at 0.802 but only GBP 636,537 of value between them - an inexpensive channel is proportionate.

Plotly: a dashboard you can email

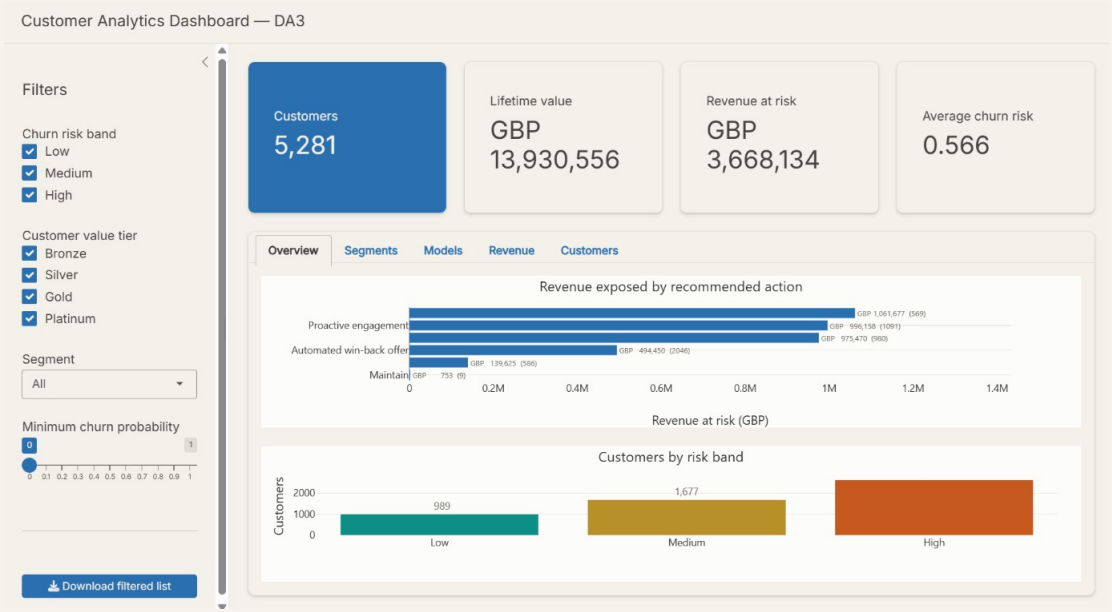


DA3_Dashboard.html - Overview tab

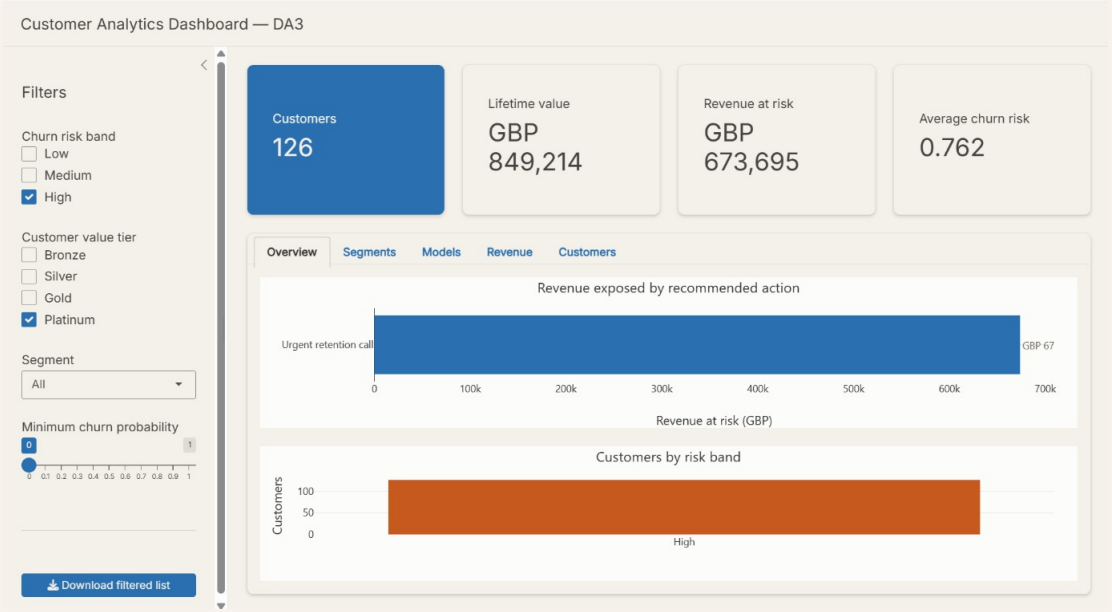
- One self-contained HTML file - the Plotly library is embedded inside it
- Opens in any browser with no server, no internet and nothing installed
- Five KPI tiles and five tabs: Overview, Model comparison, Segments, Revenue, Retention list
- Every chart is paired with a table so values can be read exactly, not estimated off an axis
- Palette checked for colour-blind separation

Shiny: reactive, not static

All filters selected



Filtered to High risk + Platinum tier



Customers	**5,281**	**126**
Revenue at risk	**GBP 3,668,134**	**GBP 673,695**
Average churn risk	**0.566**	**0.762**

Four filters feed one reactive expression, so the tiles, all six tabs and the downloadable list recalculate together.

The segments are recognisable customer types

Segment	Customers	Avg. recency	Avg. frequency	Avg. spend	Churn rate	Reads as
1	2,144	391.6 d	2.2	767	78.9%	Dormant, largely lost
2	246	32.9 d	31.4	17,069	8.9%	Loyal high spenders
3	2,876	86.7 d	5.4	1,980	44.3%	Active regular buyers
4	15	25.3 d	153.0	159,602	6.7%	Elite wholesale accounts

Clustering with no knowledge of churn

K-Means was fitted on scaled RFM values alone, yet churn rate falls from 78.9% to 6.7% straight down the table. The segments separate churn behaviour cleanly without ever being shown it - independent evidence that RFM captures something real.

Two methods, one driver

SHAP and feature importance both name Recency as the strongest predictor, with country indicators contributing almost nothing. The segment table says the same from a different angle: recency rises from 25 to 392 days and churn rises with it.

14 CONCLUSION

Four findings that outlive this dataset

1 The features set the ceiling

Five very different algorithms land between 0.7722 and 0.8070, and the top two are statistically tied. Swapping models was never going to break 0.80 - only new kinds of data would.

2 Defining the label honestly mattered most

The 90-day time-based split replaced an implausible 100% accuracy with a defensible 70 to 74%. No modelling choice moved the result nearly as much.

3 Cheap adjustments beat expensive ones

Moving the threshold to 0.453 raised recall by 0.064. A full grid search gained 0.002 of ROC-AUC. One took seconds, the other took the session.

4 Value and risk must be combined

The highest-risk customers are not the ones holding the most money. A churn model that stops at a probability column has not finished the job.

15 HONESTY

Where this stops, and what comes next

LIMITATIONS

- Churn is one 90-day window - a different window changes the labels and the accuracy
- Performance plateaus near 0.80 with no marketing, session or demographic data available
- Segment 4 holds fifteen customers, so any statistic about it is fragile
- Tuning was applied to the churn task only
- Some calibration curves deviate from the diagonal, so probabilities need calibrating before budget use
- Revenue at risk uses historical spend, and no cost of intervention is modelled

FUTURE SCOPE

- Time-dependent features - inter-purchase gaps, spend trend, seasonality
- A survival model that predicts when a customer lapses, not just whether
- Probability calibration with Platt scaling or isotonic regression
- A cost matrix, then optimise the threshold for profit rather than F1
- Extend tuning and SMOTE to the high-value and segment tasks
- Deploy the Shiny app and schedule the scoring pipeline

WHAT THREE PHASES PRODUCED

1,067,371 raw rows became a ranked retention list

5,281
customers scored

GBP 3.67M
revenue exposure quantified

569
named for urgent retention

2
dashboards delivered

Thank you

Questions welcome

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github.com/littlestuart07/Online-Retail-Analytics-Project